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B. TECH. III Sem Main 2025-2026

Computer Science and Engineering

(3CS1-01) Advanced Engineering Mathematics

Common to CS, IT, AI, DS, MC, CM, CD, CA, AD, AM, CY, IO

Time : 3 Hours]

[Max. Marks : 70

[Min. Passing Marks :

Instructions to Candidates :

Part-A : Short answer questions (up to 25 words) 10 * 2 marks = 20marks. All ten questions are compulsory.

Part-B : Analytical/Problem Solving questions 5 * 4 marks = 20 marks. Candidates have to answer 5 questions out of 7.

Part-C : Descriptive/Analytical/Problem Solving questions 3 * 10 marks = 30 marks. Candidates have to answer 3 questions out of 5.

Schematic diagrams must be shown wherever necessary. Any data you feel missing may suitably be assumed and stated clearly.

Units of quantities used/calculated must be stated clearly.

Use of following supporting materials is permitted during examination.

(Mentioned in form No. 205)

1 Normal distribution table

2 Scientific Calculator without programming

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Part-A

10x2=20

Q.1 Find the value ot k, if probability density function [2]

$$f(x) = \begin{cases} \frac{k}{\sqrt{x}} & \text{for } 0 < x < 4 \\ 0 & \text{elsewhere.} \end{cases}$$

Q.2 Define skewness of a distribution. [2]

Q.3 Write down the probability density function of Normal distribution. [2]

Q.4 If 10% of the pens manufactured by a company are defective , find the probability that a box of 12 pens contain exactly 2 defective pens. [2]

Q.5 Determine Gradient vector of $f(x) = x_1 + 2x_2 + x_1x_2 - x_1^2 - x_2^2$ [2]

Q.6 Define Non-negativity Restrictions. [2]

Q.7 Write down the types of Basic solutions. [2]

Q.8 Write down the dual problem of the primal problem : [2]

$\min z_p = C_x$

$St \;\; Ax = b$

and $x \geq 0$

Q.9 Show that : $\Delta\nabla = \delta^2 = \Delta - \nabla$ [2]

Q.10 Write down the Newton's forward difference interpolation formula. [2]

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5x4=20

Part-B

Q.1 A card is drawn from a pack of 52 cards. If the value of a card be according to its denomination viz 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 10, 10, 10, where each face card of king, queen and jack be assigned the number 10 each. Find the expected value of the number of points on the cards. [4]

Q.2 In a sample of 100 cases, the mean $\mu = 14$ and standard deviation $\sigma = 2.5$, assuming the distribution to be normal, find - [4]

(i) number of candidates who score below 10.

(ii) number of candidates whose score is between 12 and 15.

Q.3 Derive moment generating function for Binomial distribution. Hence find mean and variance. [4]

Q.4 Lines $2x + 3y = 10$ and $4x + 5y = 18$ are the lines of regressions between two variables x and y. [4]

(i) Decide which one is the line of regression y on x.

(ii) Find σx^2 if $\sigma y^2 = 9$.

Q.5 Find the volume of greates parallelopied inscribed in the ellepsoid : [4]

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

Q.6 Find a point within a triangle such that the sum of the squares of the distances from the angular points may be minimum. [4]

Q.7 Evaluate $\int_{-1.6}^{-1} e^x \, dx$ by Simpson's $\left(\frac{1}{3}\right)^{rd}$ rule with six intervals. [4]

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Part-C

3x10=30

Q.1 The first four moments of a distribution about the value 5 of a variate are 2, 20, 40 and 50. Find mean, variance, β_1 and β_2. Comment upon the nature of distribution. [10]

Q.2 Fit a second degree parabola to the following data : [10]

x	1	2	3	4	5
y	2	4	6	8	10

Q.3 Solve the following transportation problem : [10]

from ↓	← To →			available
A	16	19	12	14
B	22	13	19	16
C	14	28	8	12
Demand	10	15	17	42

Q.4 Solve the following LPP by Two phase method : [10]

$\min z = x_1 + x_2$

St $2x_1 + x_2 \geq 4$

$x_1 + 7x_2 \geq 7$

and $x_1, x_2 \geq 0$

Q.5 Solve the differential equation $\frac{dy}{dx} = x + y$, $y(0) = 1$ and obtain y for x = 0.4, the step size being 0.2. [10]

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